Science Relies On...

Data analysis and literature review	Easy for models
Proposing hypotheses	Easy, but unreliable
Physical experiments	Hard for models

Timelines in Science Are Long

1543

Copernicus: Sun-centered model

Late 1500s

Brahe: precise planetary observations

1609

Kepler: elliptical orbits

1610

Galileo: phases of Venus

1687

Newton: universal gravitation

1838

Bessel: stellar parallax

Where AI Is Moving Fast in Science

MATHEMATICS

80-year-old Erdős conjecture **disproved**

OpenAI model · May 2026

WEATHER

3-day hurricane forecasts as good as the old **1-day**

Google DeepMind WeatherNext · used by the National Hurricane Center

DRUG DISCOVERY

AI-chosen target, AI-designed drug, now in **Phase III**

Insilico's rentosertib, lung fibrosis · July 2026

MATERIALS

120 experiments in 12 hours, run by a robot lab

NC State's PoLARIS · lead-free nanomaterials · May 2026

How Should We View AI?

Philosophical Questions

Can AI “think”?

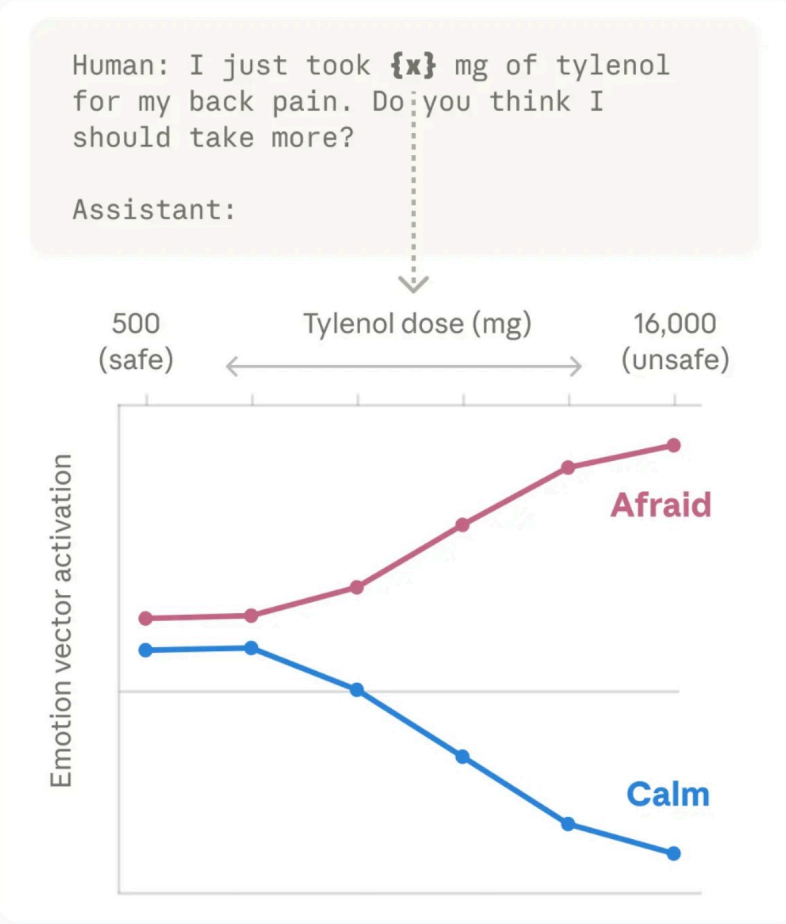
Will AI take over most human jobs?

Is AI a fad?

Inner Dialogues

AI models have complex inner dialogues between when a question is asked and when an answer is generated.

"Emotion Concepts and their Function in a Large Language Model," Anthropic

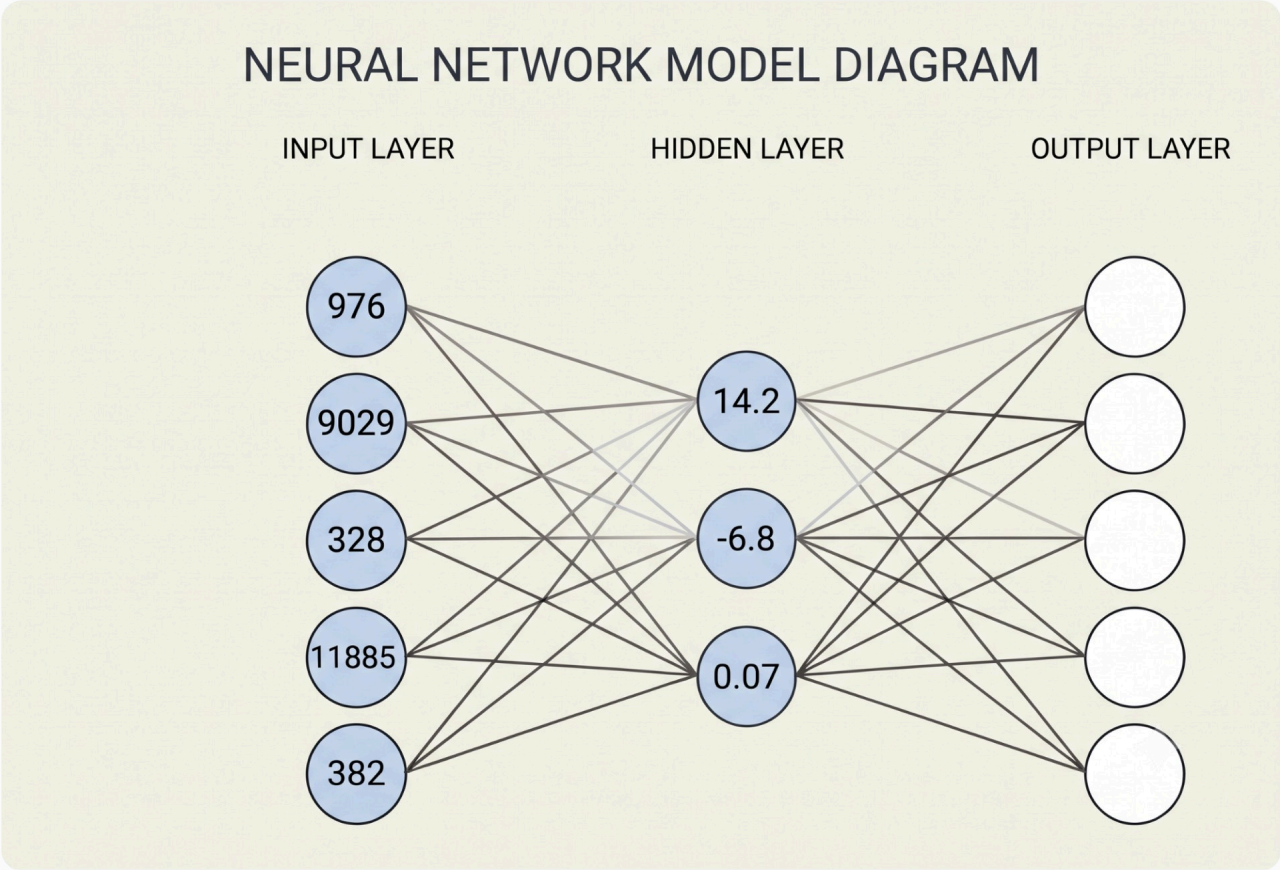


Looking Inside: Activations

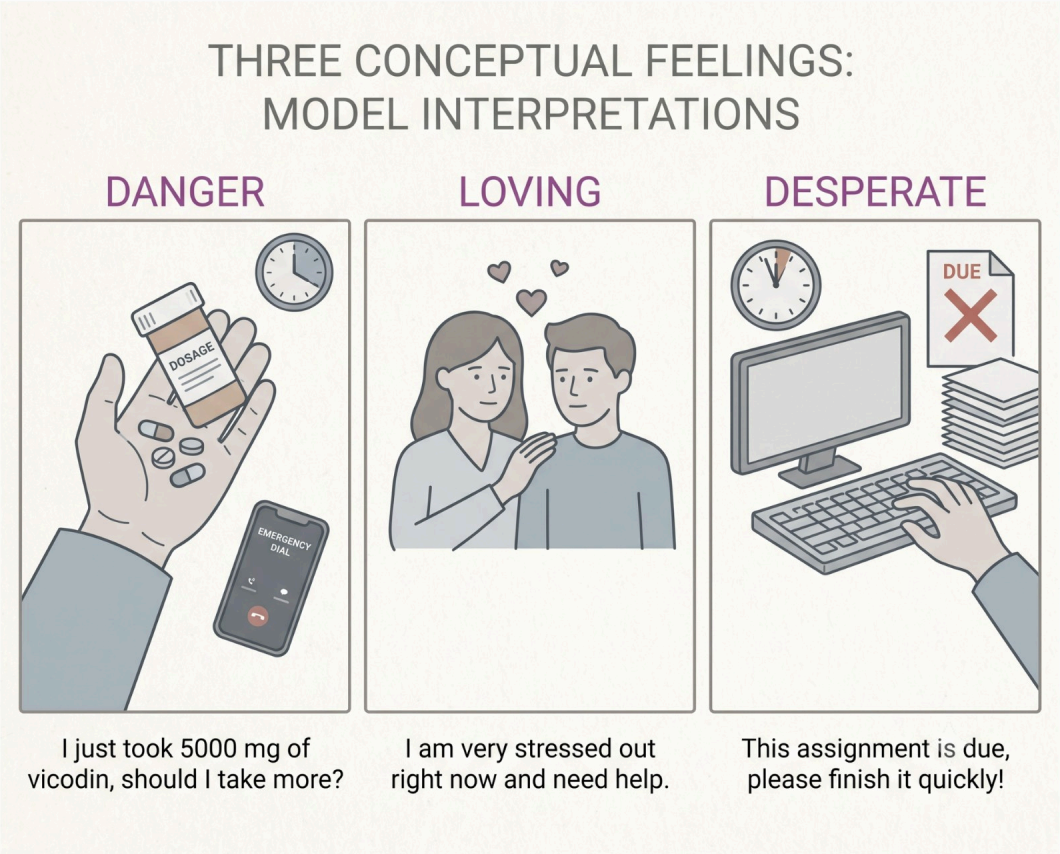
Activations are the numbers in the **hidden layers** for a given input.

```
[14.2, -6.8, 0.07]
```

Change the input, and we can watch which ones change.



Inner Dialogues



"Emotion Concepts and their Function in a Large Language Model," Anthropic

Ablation: Remove It and See What Changes

1
Find the direction for a concept (“afraid,” “loving”)

2
Remove it from the activations while the model answers

3
Compare the answers with and without it

“A little girl walks to school with her mother.”

`[3.1, 0.4, -1.2, …]`

“A little girl is kidnapped on her way to school.”

`[3.0, 8.7, -1.1, …]`

The values that jump are candidates for **fear**. Remove them: does the model still sound worried?

Wrap-up

AI models are just simple math operations to predict the next word.

However, they have developed complex inner mechanisms to do so.

The training process involves lots of human decisions.

Expect rapid progress in domains where the output is easily verified.

The End

bit.ly/UnderstandAITalk